Chapter 3
Vector Spaces

3.1 Vector and Vector Space Axioms

Logic of Chapter 3

$A\boldsymbol{x} = \boldsymbol{b}$

$x_1\boldsymbol{a}_1 + x_2\boldsymbol{a}_2 + \cdots + x_n\,\boldsymbol{a}_n = \boldsymbol{b}$ Ch3.3 & Ch3.4

Consistency

Consistency: $\boldsymbol{b}$ in the **column space of** $\boldsymbol{A}$: $col\ \boldsymbol{A}$ Ch3.6

Row echelon form

What the solution set looks like: forms a **subspace.** Ch3.2

Invertibility of A

How large is the solution set
basis, dimension, rank,
The Rank–Nullity Theorem,
$\text{rank}(A) + \dim N(A) = n$ Ch3.4 Ch3.6

$Det(A)$

Are there solutions?
How to find all solutions?

Changing basis Ch3.5

Objective

- n-dimensional real vector space
  - 二维、三维空间是最经典的向量空间：向量可以“加减”和“缩放”
- Vector Space Axioms
  - 向量空间的抽象定义，只需定义“向量加法”和“标量乘法”，并且满足一系列公理，就是向量空间
- Other examples of vector spaces
  - “矩阵”、“函数”、“多项式”等也可以形成向量空间，因为加减法、标量乘法满足公理

R^n $R^{m\times n}$ $C[a,b]$ P_n

$\mathbb{R}^n$

Definition

The set of all column n-vectors $[u_1, u_2, \dots, u_n]^T$ is called ***n-dimensional real space*** and is denoted $\mathbb{R}^n$.

u_i is the ***i*-th component** of $[u_1, u_2, \dots, u_n]^T$.

$u_i \in \mathbb{R}$

Example

$\mathbb{R}^4$ is the sets of all 4x1 column vectors. For example, $[1, 2, 3, 4]^T$ and $[-1, 3, 5.2, 0]^T$ are in $\mathbb{R}^4$.

$\mathbb{R}^5$ is the set of all 5x1 column vectors. For example, $[-1, 2, 0, 3, 9]^T$ is in this set.

Addition and Scalar Multiplication

Definition

Let $\mathbf{u} = [u_1, u_2, \ldots, u_n]^T$ and $\mathbf{v} = [v_1, v_2, \ldots, v_n]^T$ be two elements of $\mathbb{R}^n$. We say that **u** and **v** are **equal** if $u_1 = v_1$, …, $u_n = v_n$.

Let $\mathbf{u} = [u_1, u_2, \ldots, u_n]^T$ and $\mathbf{v} = [v_1, v_2, \ldots, v_n]^T$ be two elements of $\mathbb{R}^n$ and let c be a scalar. Addition and scalar multiplication are defined as

加法

Addition: $\mathbf{u} + \mathbf{v} = [u_1 + v_1, u_2 + v_2, \ldots, u_n + v_n]^T \in \mathbb{R}^n$

标量乘法

Scalar multiplication : $c\mathbf{u} = [cu_1, cu_2, \ldots, cu_n]^T \in \mathbb{R}^n$

Note.

在加法下闭合

(1) **u**, **v**$\in \mathbb{R}^n \Rightarrow$ **u**+**v** $\in \mathbb{R}^n$ ($\mathbb{R}^n$ is **closed** under addition)

(2) **u**$\in \mathbb{R}^n$, $c \in$ **R** $\Rightarrow c$**u** $\in \mathbb{R}^n$ ($\mathbb{R}^n$ is **closed** under scalar multiplication)

在标量乘法下闭合

Let V

(1) $\forall x, y \in V$ $\quad x+y \in V$

V is closed under addition.

(2) $\forall x \in V$ $\quad \forall \alpha \in R$

$\alpha x \in V$, V is closed under scalar multiplication

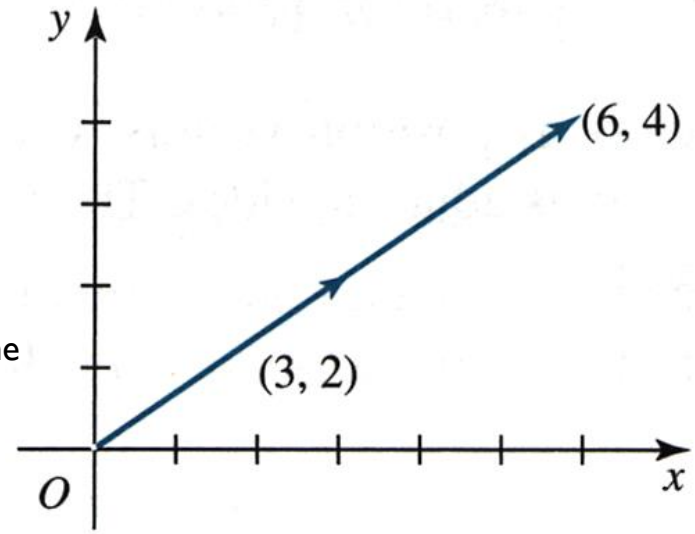
Scalar multiplication example

Consider the scalar multiple of the vector $[3,2]^T$ by 2, we get

$$2[3,2]^T = [6,4]^T$$

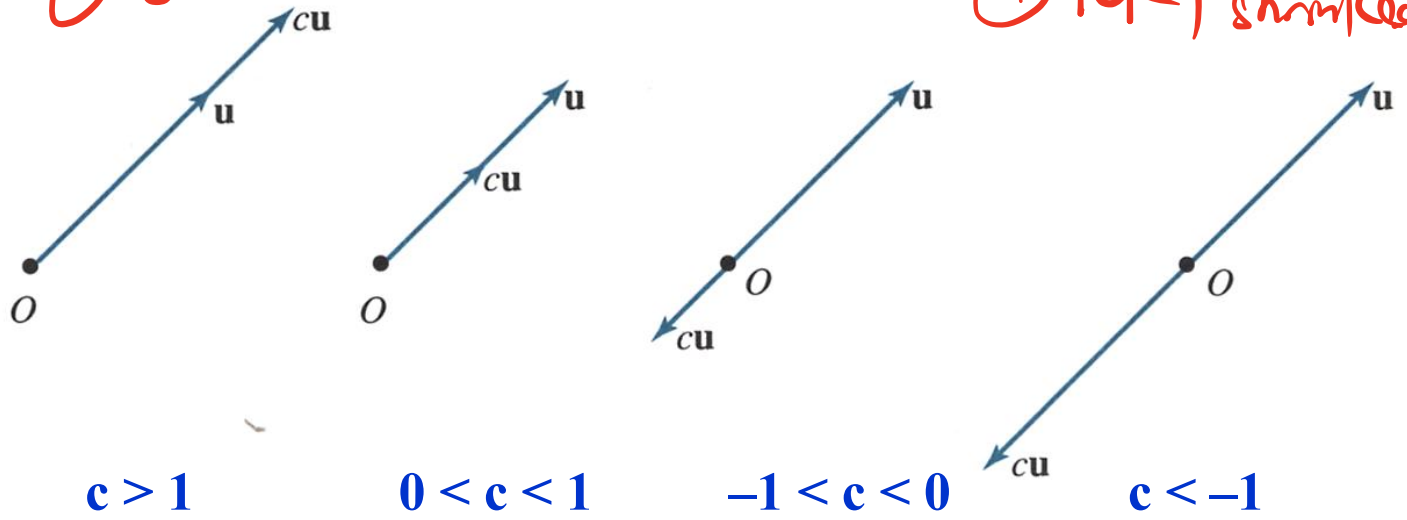
Observe that $[6,4]^T$ is a vector in the same direction as $[3,2]^T$, but 2 times further away.

(Distracting question on the side: what is the "length" of the vector $[3,2]^T$? How should you calculate the "length" of an n-vector? More about that in the future.)



① $c > 0$ same direction ③ $|c| > 1$ enlarge

② $c < 0$ opposite direction ④ $|c| < 1$ shrink



$c > 1$ $0 < c < 1$ $-1 < c < 0$ $c < -1$

- The direction of $c\mathbf{u}$ will be
  - the same as the direction of **u** if $c > 0$,
  - the opposite direction to **u** if $c < 0$.
- The "length" of $c\mathbf{u}$ is $|c|$ times the length of **u**.
- Property: multiplying **u** by β times, and then by α times, is the same as multiplying **u** by $\alpha \times \beta$ times.

$\alpha(\beta u) = (\alpha\beta)u = \beta(\alpha u)$

Zero vector, negative vector

The vector $[0,0,\ldots,0]^T$, having *n* zero components, is called the **zero vector** of $\mathbb{R}^n$ and is denoted **0**.

Property: for any vector v, **0 + v = v, v + 0 = v.**

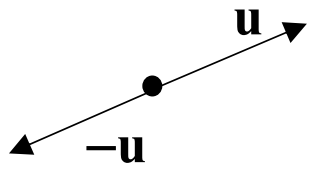
$\in \mathbb{R}^n$ $\in \mathbb{R}^n$

Negative Vector

The vector **(–1)u** is written **–u** and is called the **negative of u**. It is a vector having the same magnitude as **u**, but lies in the opposite direction to **u**.

Property: for any vector **u**,

u + (–u) = 0 = (- u) + u



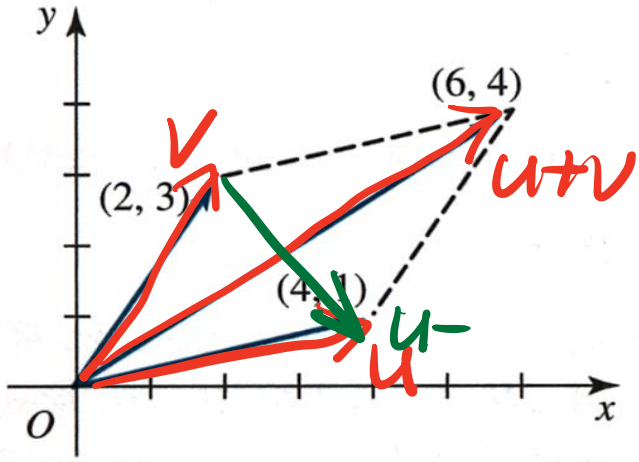
$\forall u \in \mathbb{R}^n$ $-u \in \mathbb{R}^n$

Addition and subtraction

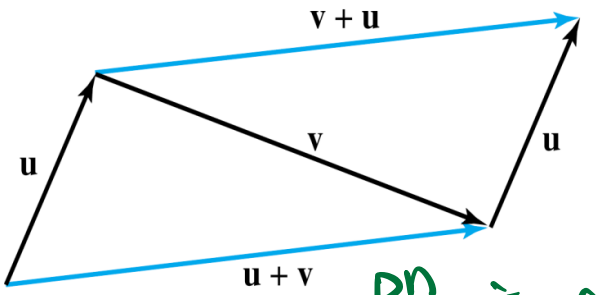
- if **u** and **v** are 2 vectors, then **u+v** is another vector:
  - Algebraically, **u+v** = $\begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$
  - Geometrically, **u** + **v** is the diagonal of the parallelogram（平行四边形） defined by **u** and **v**.
- $\mathbf{u} - \mathbf{v}$ is the vector pointing from $\mathbf{v}$ to $\mathbf{u}$.

Example: Consider the vector (4, 1 and (2, 3), then

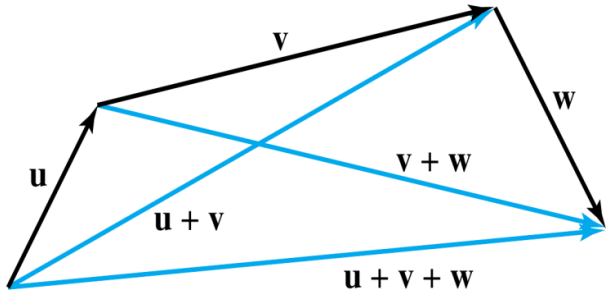
$$\begin{bmatrix} 4 \\ 1 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}.$$



Properties of vector additions



$\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}.$



$(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w}).$

R^n is closed under addition and scalar multiplication.

Vector addition and scalar multiplications have many good properties:

A1. **x** + **y** = **y** + **x** for $\forall\, x, y \in V$

A2. (**x** + **y**) + **z** = **x** + (**y** + **z**), for all **x, y, z** in V

A3. There exist an element **0** $\in V$, such that **x** + **0** = **0** + **x** = **x** holds for any **x**

A4. for any x, there exist an (negative) element, -**x**, such that x + (–**x**) = **0**

A5. α(**x** + **y**) = α**x** + α**y**

A6. (α + β) **x** = α**x** + β **x**

A7. ($\alpha\beta$)**x** = α(β**x**)

A8. 1 **x** = **x**

In R^n $-x = (-1)x$

Question: where else have you seen similar properties of addition and scalar multiplication?

Vector Space – Abstract definition

Definition

Let V be a set on which the operations of *addition* and *scalar multiplication* are defined. By this we mean that,

C1 with each pair of elements **x** and **y** in V, there exists a unique element **x** + **y** that is **also in** $\boldsymbol{V}$,

C2 with each element **x** in V and each scalar α, there exists a unique element $\alpha \circ \mathbf{x}$ that is **also in** $\boldsymbol{V}$.

Then V together with operations addition and scalar multiplication is called a ***vector space*** if the following axioms are satisfied (A1-A8):

A1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$ for $\forall\, x, y \in V$

A2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$, for all **x, y, z** in V

A3. There exist an element $\mathbf{0} \in V$, such that $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$ holds for any **x**

A4. for any x, there exist an (negative) element, -**x**, such that $x + (-\mathbf{x}) = \mathbf{0}$

A5. $\alpha \circ (\mathbf{x} + \mathbf{y}) = \alpha \circ \mathbf{x} + \alpha \circ \mathbf{y}$

A6. $(\alpha + \beta) \circ \mathbf{x} = \alpha \circ \mathbf{x} + \beta \circ \mathbf{x}$

A7. $(\alpha\beta) \circ \mathbf{x} = \alpha \circ (\beta \circ \mathbf{x})$

A8. $1 \circ \mathbf{x} = \mathbf{x}$

C1: $\forall x, y \in V \quad x+y \in V$

C2: $\forall x \in V, \forall \alpha \in \mathbb{R}, \quad \alpha x \in V$ > closure properties.

A1: $\forall x, y \in V \quad x+y = y+x$

A2: $\forall x, y, z \in V \quad (x+y)+z = x+(y+z)$.

A3: $\exists$ "0" $\in V$ such that $\forall x \in V$

$x + "0" = "0" + x = x$

T① "0" is a name. "0" $\neq [0, 0, \cdots, 0]^T$

If $V \neq \mathbb{R}^n$

② You need to show "0" exists.

You need to give the determined expression of "0"

A4: $\exists$ "$-x$" $\in V$ such that $\forall x \in V$

$x + "-x" =$ "0" → zero-vector you found in A3.

A5: $\forall \alpha \in \mathbb{R} \quad \forall x, y \in V$

$\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y$

defined scalar multiplication

defined addition

A6 $\forall \alpha, \beta \in \mathbb{R} \quad \forall x \in V$

$(\alpha + \beta) \cdot x = \alpha \cdot x + \beta \cdot x$

normal addition

A7: $\forall \alpha, \beta \in \mathbb{R} \quad \forall x \in V$

$(\alpha\beta) \cdot x = \alpha \cdot (\beta \cdot x)$

normal

defined

defined

A8. $\forall x \in V$

$1 \cdot x = x$

real number

Review: Matrix Operations

We have seen something similar to the 8 axioms of vector spaces when we defined matrix addition and scalar multiplication in Section 1.3.

Theorem

Let A, B, and C be matrices and a, b, and c be scalars. Assume that the size of the matrices are such that the operations can be performed.

Properties of Matrix Addition and scalar Multiplication

1. $A + B = B + A$ — *Commutative property of addition*
2. $A + (B + C) = (A + B) + C$ — *Associative property of addition*
3. $A + O = O + A = A$ — (*where O is the zero matrix*)
4. $A + (-A) = 0$
5. $c(A + B) = cA + cB$ — *Distributive property of addition*
6. $(a + b)C = aC + bC$ — *Distributive property of addition*
7. $(ab)C = a(bC)$
8. $1C = C$

$\mathbb{R}^{m\times n}$ is a Vector Space

matrix space.

- $\mathbb{R}^{m\times n}$ is the set of all $m \times n$ matrices. (So, here a "vector" is one $m \times n$ matrix)
- Addition: $A + B = (a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij})$ [the sum is still an $m \times n$ matrix]
- Scalar multiplication: $cA = c(a_{ij}) = (ca_{ij})$. [It is still an $m \times n$ matrix]
- The axioms and closure property hold, and therefore $\mathbb{R}^{m\times n}$, with matrix addition
and scalar multiplication, is a vector space.

1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$ $\quad (a_{ij}) + (b_{ij}) = (a_{ij} + b_{ij})$
2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$ $\quad \big((a_{ij}) + (b_{ij})\big) + (c_{ij}) = (a_{ij}) + \big((b_{ij}) + (c_{ij})\big)$
3. $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$ $\quad (a_{ij}) + (0) = (0) + (a_{ij}) = (a_{ij})$
4. $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$ $\quad (a_{ij}) + (-a_{ij}) = (0)$
5. $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\,\mathbf{x} + \alpha\,\mathbf{y}$ $\quad \alpha\big((a_{ij}) + (b_{ij})\big) = \alpha\,(a_{ij}) + \alpha(b_{ij})$
6. $(\alpha + \beta)\,\mathbf{x} = \alpha\,\mathbf{x} + \beta\,\mathbf{x}$ $\quad (\alpha + \beta)(a_{ij}) = \alpha\,(a_{ij}) + \beta(b_{ij})$
7. $\alpha\beta(\mathbf{x}) = (\alpha\beta)\mathbf{x}$ $\quad (\alpha\beta)(a_{ij}) = \alpha(\beta(a_{ij}))$
8. $1\,\mathbf{x} = \mathbf{x}$ $\quad 1\,(a_{ij}) = (a_{ij})$

C1: $\forall A, B \in R^{m\times n}$

$\underbrace{A+B}_{m\times n} \in R^{m\times n}$

C2: $\forall A \in R^{m\times n} \quad \forall \alpha \in R$

$\underbrace{\alpha A}_{m\times n} \in R^{m\times n}$

A1-A8.

A3: "0" $= O_{m\times n} \in R^{m\times n}$

$$O = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

$\forall A \in R^{m\times n} \quad A+O=O+A=A$

A4: "$-A$" $= (-1)A \quad \forall A \in R^{m\times n}$

$A+$"$-A$" $= A+(-1)A = \underbrace{O}_{} =$ "0"

$C[a, b]$ is a Vector Space

- $C[a, b]$ is the set of all real-valued continuous functions defined on the interval $[a, b]$. Vectors in this case are continuous real-valued functions.
- For $f, g \in C[a, b]$ and scalar α, define
  - addition: $(f + g)(x) = f(x) + g(x)$
  - scalar multiplication: $(\alpha f)(x) = \alpha f(x)$.
- Check that axioms and closure property hold.

C1. $\alpha \mathbf{x} \in V$.
C2. $\mathbf{x} + \mathbf{y} \in V$.
A1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
A2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$
A3. $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$
A4. $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$

A5. $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\,\mathbf{x} + \alpha\,\mathbf{y}$
A6. $(\alpha + \beta)\,\mathbf{x} = \alpha\,\mathbf{x} + \beta\,\mathbf{x}$
A7. $\alpha\beta(\mathbf{x}) = (\alpha\beta)\mathbf{x}$
A8. $1\,\mathbf{x} = \mathbf{x}$

C1: $\forall f \in C[a,b]$ $\forall \alpha \in \mathbb{R}$

$(\alpha f)(x) = \alpha f(x) \in C[a,b]$

Since $\alpha f(x)$ is still a continuous function on $[a,b]$

C2: $\forall f, g \in C[a,b]$

$(f+g)(x) = f(x) + g(x) \in C[a,b]$

since the addition between continuous functions is still continuous.

A1: $\forall f, g \in C[a,b]$

$(f+g)(x) = f(x) + g(x) = g(x) + f(x) = (g+f)(x)$

A2: $\forall f, g, h \in C[a,b]$

$$[(f+g)+h](x) = (f+g)(x) + h(x)$$
$$= f(x) + g(x) + h(x)$$
$$= f(x) + [(g+h)(x)$$

$= [f+(g+h)](x)$

A3. $\exists$ "0" $\in V$ such that $\forall x \in V$

$x + "0" = x$

"0"

Define $z(x)=0$ $x \in [a,b]$ $\in C[a,b]$

$\forall f \in C[a,b]$

$(f+z)(x) = f(x)+z(x)$

$= f(x)$

A4. $\exists$ "$-f$" $\in C[a,b]$ such that

$\forall f \in C[a,b]$

$f + "-f" = "0"$

"$-f$" $= (-1)f(x)$

$\forall f \in C[a,b]$

$(f + "-f")(x) = f(x) + "-f"(x)$

$= f(x) + (-1)f(x) = 0 = z(x)$

"0"

P_n is a Vector Space

- **Definition:** a polynomial of degree k $(k \geq 0)$ is a function of the form

$p(x) = a_0 + a_1x + a_2x^2 + \ldots + a_kx^k$ where $a_k \neq 0$.

0(x) = 0 is also considered to be a polynomial.

- Example:
  - Polynomials of degree 0 (non-zero constants) 1, 2, -3
  - Polynomials of degree 1 (linear functions): x, 2x-1, 3x+8,-100x
  - Polynomials of degree 2 (quadratic functions): $x^2, 2x^2, 3x^2 + 2x + 1$
- **Definition:** P_n is the set of all polynomials of degree **strictly less than n**. Vectors in this case are polynomial functions of the form $p(x) = a_0 + a_1x + a_2x^2 + \ldots + a_{n-1}x^{n-1}$. For $p, q \in P_n$ and scalar α, define addition by $(p + q)(x) = p(x) + q(x)$ and scalar multiplication by $(\alpha p)(x) = \alpha p(x)$.

$p(x) = 1 + 2x^2 + -x^4$ → degree is 4.

$p(x) = 1 + 2x^2 + 0x^4$

$= 1 + 2x^2$ → degree is 2.

P_4: is a set contains all polynomials with degree less than 4.

≤ 3.

$p(x) \in P_4$

$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$

Eg. $x + 2 \in P_4$

$x^3 + 2x^2 + 6 \in P_4$

P_n is a Vector Space

- Definition: P_n is the set of all polynomials of degree **strictly less than n**. Vectors in this case are polynomial functions of the form $p(x) = a_0 + a_1x + a_2x^2 + \ldots + a_{n-1}x^{n-1}$. For $p, q \in P_n$ and scalar α, define addition by $(p + q)(x) = p(x) + q(x)$ and scalar multiplication by $(\alpha p)(x) = \alpha p(x)$.
- Check that axioms and theorem hold.

C1. $\alpha \mathbf{x} \in V$.

C2. $\mathbf{x} + \mathbf{y} \in V$.

A1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$

A2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$

A3. $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$

A4. $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$

A5. $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\,\mathbf{x} + \alpha\,\mathbf{y}$

A6. $(\alpha + \beta)\,\mathbf{x} = \alpha\,\mathbf{x} + \beta\,\mathbf{x}$

A7. $\alpha\beta(\mathbf{x}) = (\alpha\beta)\mathbf{x}$

A8. $1\,\mathbf{x} = \mathbf{x}$

C1: $\forall \alpha \in R \quad \forall p \in P_n$

$(\alpha p)(x) = \alpha p(x) = \alpha(a_0 + a_1 x + \cdots + a_{n-1}x^{n-1})$

$= (\alpha a_0) + (\alpha a_1)x + \cdots + (\alpha a_{n-1})x^{n-1} \in P_n$

C2: $\forall\ p, q \in P_n$

$p(x) = a_0 + a_1 x + \cdots + a_{n-1}x^{n-1}$

$q(x) = b_0 + b_1 x + \cdots + b_{n-1}x^{n-1}$

$(p+q)(x) = p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \cdots$

$+ (a_{n-1} + b_{n-1})x^{n-1} \in P_n$

A1: $\forall\ p, q \in P_n$

$(p+q)(x) = p(x) + q(x) = q(x) + p(x) = (q+p)(x)$.

A3: $\exists$ $0(x) = 0 \in P_n$

is a polynomial with degree 0.

$\forall\ p \in P_n$

$$(p+0)(x) = p(x) + 0(x) = p(x) + 0 = p(x)$$

Aft: For $\forall\, p \in P_n$

$\exists$ "$-p$" $= (-1)p(x)$.

$$(p + \text{"}-p\text{"})(x) = p(x) + (-1)p(x) = 0 = 0(x)$$

Commonly used vector spaces.

$\mathbb{R}^n$ $\quad$ $\mathbb{R}^{m\times n}$ $\quad$ $C[a,b]$ $\quad$ P_n

$\mathbb{R}^n$

↑

$\mathbb{R}^{1\times n}$

Example: NOT a Vector Space

- The set V of all polynomials of degree **exactly** 3, with the usual addition and scalar multiplication, **IS NOT** a vector space. Which of the closure properties and axioms fails?
- Axioms that hold: A1, A2, A5, A6, A7, A8
- Axioms that fail:
  - The zero polynomial 0(x) does not have degree 3, so 0(x) is not in the set V. So A3 and A4 fails.
  - C1 fails because if α=0, then αx^3=0 which is not in V.
  - C2 fails because x^3 and $-x^3 + x^2$ are both in V, but $x^3 + \left(-x^3 + x^2\right) = x^2$ has degree 2, so it is not in V.

C1. $\alpha \mathbf{x} \in V$.
C2. $\mathbf{x} + \mathbf{y} \in V$.
A1. $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$
A2. $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$
A3. $\mathbf{x} + \mathbf{0} = \mathbf{0} + \mathbf{x} = \mathbf{x}$
A4. $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$

A5. $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\,\mathbf{x} + \alpha\,\mathbf{y}$
A6. $(\alpha + \beta)\,\mathbf{x} = \alpha\,\mathbf{x} + \beta\,\mathbf{x}$
A7. $\alpha\beta(\mathbf{x}) = (\alpha\beta)\mathbf{x}$
A8. $1\,\mathbf{x} = \mathbf{x}$

$\forall p \in V \qquad p(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3$

$a_3 \neq 0$

Cl: $[\ \forall \alpha \in R \ , \ \forall x \in V \qquad \alpha x \in V\ ]$

let $\alpha = 0$ $\qquad \forall p \in V$

$(\alpha p)(x) = \alpha p(x) = 0$ $\qquad$ degree $= 0 \notin V$

V is not closed under scalar multip—

$\Rightarrow$ V is not a vector space.

Theorem

If V is a vector space and $\mathbf{x} \in V$, then

1. $0\mathbf{x} = \mathbf{0}$
2. $\mathbf{x} + \mathbf{v} = \mathbf{0}$ implies that $\mathbf{v} = -\mathbf{x}$
3. $(-1)\mathbf{x} = -\mathbf{x}$.

Proof

1. x=(1+0)x=1x+0x=x+0x, therefore 0x=0
2. x+v=0, then –x=-x+0=-x+(x+v)=v
3. 0=0x=(1-1)x=1x+(-1)x, x+(-1)x=0, therefore (-1)x=-x

$0, 1$: scalar.

$\underline{x}$ $\underline{0}$: vectors.

(1) $0 \cdot \underline{x} = \underline{0}$

⊤ "$\underline{0}$" $\in V$

$\underline{x} + \text{"}\underline{0}\text{"} = \underline{x}$

$$\underline{x} = 1 \cdot \underline{x} = (1+0) \cdot \underline{x} = 1 \cdot \underline{x} + 0 \cdot \underline{x} = \underline{x} + 0 \cdot \underline{x}$$

A8 A6

$\Rightarrow \underline{x} + 0\underline{x} = \underline{x}$

$\Rightarrow 0\underline{x} = \text{"}\underline{0}\text{"}$

⊤ If $p \rightarrow q$.

If not $q \rightarrow$ not p

(2) $$-\underline{x} = -\underline{x} + \underline{0} = -\underline{x} + (\underline{x} + \underline{v}) = (-\underline{x} + \underline{x}) + \underline{v}$$

A3 A2 A4

$$= \underline{0} + \underline{v} = \underline{v} \Rightarrow \underline{v} = -\underline{x}$$

A3

(3) $\underline{0} = 0 \cdot \underline{x} = (1-1) \cdot \underline{x} = 1 \cdot \underline{x} + (-1)\underline{x}$

$= \underline{x} + (-1)\underline{x}$

$\underline{x} + (-1)\underline{x} = \underline{0}$

$\Rightarrow (-1)\underline{x} = -\underline{x}$

$\underline{x} + \underline{v} = \underline{0} \rightarrow \underline{v} = -\underline{x}$

Let R^+ denote the set of positive real numbers. Define the operation of scalar multiplication, denoted $\circ$, by

$$\alpha \circ x = x^{\alpha}$$

For each $x \in R^+$ and for any real number α. Define the operation of addition, denoted $\oplus$, by

$$x \oplus y = x \cdot y \text{ for all } x, y \in R^+$$

Thus for this system the scalar product of -3 times $\frac{1}{2}$ is given by

$$-3 \circ \frac{1}{2} = \left(\frac{1}{2}\right)^{-3} = 8$$

And the sum of 2 and 5 is given by

$$2 \oplus 5 = 2 \cdot 5 = 10$$

Is R^+ a vector space with these operations? Prove your answer.

C1: $\forall x, y \in R^+$ $\quad x \oplus y = x \cdot y \in R^+$

positive real number.

C2: $\forall \alpha \in R \quad \forall x \in R^+$

$\alpha \circ x = x^\alpha \in R^+$

positive real number.

A1: $\forall x, y \in R^+$

$x \oplus y = x \cdot y = y \cdot x = y \oplus x$

A2: $\forall x, y, z \in R^+$

$(x \oplus y) \oplus z = (x \cdot y) \oplus z = (x \cdot y) \cdot z$

$= x \cdot (y \cdot z) = x \oplus (y \oplus z)$

A3 "0" A4 "-x" → Write out the detailed expression of "0", "-x".

If V is a vector space. "0" = 0 ∘ x

"-x" = (-1) ∘ x

let "0" = 0 ∘ x if x + "0" = x A3 satisfies.

If x + "0" ≠ x . "0" doesn't exist. V is not a vector space

A3: $\forall x \in \mathbb{R}^+$ let $"0" = 0 \cdot x = x^0 = 1 \in \mathbb{R}^+$

[A3: $\exists "0" \in V$ such that $\forall x \in V$ $x + "0" = x$]

$x \oplus "0" = x \oplus 1 = x \cdot 1 = x$

A4. $\forall x \in \mathbb{R}^+$ let $"-x" = (-1) \circ x = x^{-1} = \frac{1}{x} \in \mathbb{R}^+$

$x \oplus "-x" = x \oplus \frac{1}{x} = x \cdot \frac{1}{x} = 1 = "0"$

A5: $\forall \alpha \in \mathbb{R}$ $\forall x, y \in \mathbb{R}^+$

$\alpha \circ (x \oplus y) = \alpha \circ (x \cdot y) = (x \cdot y)^\alpha = x^\alpha y^\alpha$

$(\alpha \circ x) \oplus (\alpha \circ y) = x^\alpha \oplus y^\alpha = x^\alpha y^\alpha$ //

A6: $\forall \alpha, \beta \in \mathbb{R}$ $\forall x \in \mathbb{R}^+$

$(\alpha + \beta) \circ x = x^{\alpha+\beta} = x^\alpha \cdot x^\beta$

$+$ ↗ normal scalar addition

$(\alpha \circ x) \oplus (\beta \circ x) = x^\alpha \oplus x^\beta = x^\alpha \cdot x^\beta$ //

A7: $\forall \alpha, \beta \in \mathbb{R}$ $\forall x \in \mathbb{R}^+$

$(\alpha\beta) \circ x = x^{\alpha\beta}$

normal

$\alpha \circ (\beta \circ x) = \alpha \circ x^{\beta} = (x^{\beta})^{\alpha} = x^{\alpha\beta}$

AB $\forall x \in \mathbb{R}^{+}$

$1 \circ x = x^{1} = x$

Yes. Proof:

C1. Let $x, y \in \Re^+$. Then $x \cdot y$ are positive real numbers. So $x \oplus y = x \cdot y$ is a positive real number and is in $\Re^+$.

C2. Let $x \in \Re^+$ and $\alpha \in \boldsymbol{R}$. Then x is a positive real number. So $\alpha \circ x = x^\alpha$ is a positive real number and is in $\Re^+$.

A1. $x \oplus y = x \cdot y = y \cdot x = y \oplus x$

A2. $(x \oplus y) \oplus z = x \cdot y \cdot z = x \oplus (y \oplus z)$

A3. Since $x \oplus 1 = x \cdot 1 = x$ for all x, it follows that 1 is the zero vector.

A4. Let

$$-x = -1 \circ x = x^{-1} = \frac{1}{x}$$

It follows that

$$x \oplus (-x) = x \cdot \frac{1}{x} = 1 \quad \text{(the zero vector)}.$$

Therefore $\frac{1}{x}$ is the additive inverse of x for the operation $\oplus$.

A5. $\alpha \circ (x \oplus y) = (x \oplus y)^\alpha = (x \cdot y)^\alpha = x^\alpha \cdot y^\alpha$

$\alpha \circ x \oplus \alpha \circ y = x^\alpha \oplus y^\alpha = x^\alpha \cdot y^\alpha$

A6. $(\alpha + \beta) \circ x = x^{(\alpha+\beta)} = x^\alpha \cdot x^\beta$

$\alpha \circ x \oplus \beta \circ x = x^\alpha \oplus x^\beta = x^\alpha \cdot x^\beta$

A7. $(\alpha\beta) \circ x = x^{\alpha\beta}$

$\alpha \circ (\beta \circ x) = \alpha \circ x^\beta = (x^\beta)^\alpha = x^{\alpha\beta}$

A8. $1 \circ x = x^1 = x$

Since all eight axioms hold, R^+ is a vector space under the operations of $\circ$ and $\oplus$.

Summary

- Vector spaces are defined to be sets with addition and scalar multiplication, satisfying closure and other 8 axioms.
- To see if a set forms a vector space under addition and scalar multiplication, check 2 closure properties and 8 axioms.
- Besides the vector space of $\mathbb{R}^n$, many useful sets are vector spaces, including
  - The set of all matrices of size $m \times n$
  - The set of all continuous functions defined on [a, b]
  - The set of all polynomials with degree $< n$.